# SEMESTER 5

# MACHINE LEARNING LAB

(Common to CS/CA)

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| **Course Code** | **PCCSL508** | **CIE Marks** | 50 |
| **Teaching Hours/Week (L: T:P: R)** | 0:0:3:0 | **ESE Marks** | 50 |
| **Credits** | 2 | **Exam Hours** | 2 Hrs. 30 Min. |
| **Prerequisites (if any)** | None | **Course Type** | Lab |

**Course Objectives:**

1. To give the learner a practical experience of the various machine learning techniques and be able to demonstrate them using a language of choice.

| **Expt. No.** | **Experiments** |
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| 1 | Implement linear regression with one variable on the California Housing dataset to predict housing prices based on a single feature (e.g., the average number of rooms per dwelling).<br>Tasks:<br>● Load and preprocess the datase.<br>● Implement linear regression using both gradient descent and the normal equation.<br>● Evaluate the model performance using metrics such as Mean Squared Error (MSE) and R-squared.<br>● Visualize the fitted line along with the data points. |
| 2 | Implement polynomial regression on the Auto MPG dataset to predict miles per gallon (MPG) based on engine displacement. Compare polynomial regression results with linear regression.<br>Tasks:<br>● Load and preprocess the dataset.<br>● Implement polynomial regression of varying degrees.<br>● Compare the polynomial regression models with linear regression using metrics such as MSE and R-squared.<br>● Visualize the polynomial fit. |
| 3 | Implement Ridge and Lasso regression on the Diabetes dataset. Compare the performance of these regularized models with standard linear regression.<br>Tasks: |

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| | ● Load and preprocess the dataset.<br>● Implement Ridge and Lasso regression.<br>● Tune hyperparameters using cross-validation.<br>● Compare performance metrics (MSE, R-squared) with standard linear regression. |
| 4 | Estimate the parameters of a logistic regression model using MLE and MAP on the Breast Cancer Wisconsin dataset. Compare the results and discuss the effects of regularization.<br>Tasks:<br>● Load and preprocess the dataset.<br>● Implement logistic regression with MLE.<br>● Apply MAP estimation with different regularization priors (L1 and L2 regularization).<br>● Compare the performance and parameter estimates with MLE and MAP. |
| 5 | Use MLE and MAP to estimate the parameters of a multinomial distribution on the 20 Newsgroups dataset. Explore the impact of different priors on the estimation.<br>Tasks:<br>● Load and preprocess the dataset.<br>● Implement MLE for multinomial distribution parameter estimation.<br>● Apply MAP estimation with various priors (e.g., Dirichlet priors).<br>● Compare results and evaluate the effect of different priors. |
| 6 | Implement a logistic regression model to predict the likelihood of a disease using the Pima Indians Diabetes dataset. Compare the performance with and without feature scaling.<br>Tasks:<br>● Load and preprocess the Pima Indians Diabetes dataset.<br>● Implement logistic regression for binary classification.<br>● Evaluate model performance with and without feature scaling.<br>● Analyze metrics such as accuracy, precision, recall, and F1-score. |
| 7 | Implement a Naïve Bayes classifier to categorize text documents into topics using the 20 Newsgroups dataset. Compare the performance of Multinomial Naïve Bayes with Bernoulli Naïve Bayes.<br>Tasks:<br>● Load and preprocess the 20 Newsgroups dataset.<br>● Implement Multinomial Naïve Bayes and Bernoulli Naïve Bayes classifiers.<br>● Evaluate and compare the performance of both models using metrics such as accuracy and F1-score.<br>● Discuss the strengths and weaknesses of each Naïve Bayes variant for text classification. |

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| 8 | Implement the K-Nearest Neighbors (KNN) algorithm for image classification using the Fashion MNIST dataset. Experiment with different values of K and analyze their impact on model performance.<br>Tasks:<br>● Load and preprocess the Fashion MNIST dataset.<br>● Implement KNN for multi-class classification.<br>● Experiment with different values of K and evaluate performance.<br>● Discuss the impact of different K values on model accuracy and computational efficiency. |
| 9 | Implement a Decision Tree classifier using the ID3 algorithm to segment customers based on their purchasing behavior using the Online Retail dataset. Analyze the tree structure and discuss the feature importance.<br>Tasks:<br>● Load and preprocess the Online Retail dataset.<br>● Implement Decision Tree using the ID3 algorithm.<br>● Visualize the decision tree and analyze feature importance.<br>● Discuss how the tree structure helps in understanding customer behavior. |
| 10 | Implement and compare Logistic Regression and Decision Trees on the Adult Income dataset for predicting income levels. Evaluate both models based on performance metrics and interpretability.<br>Tasks:<br>● Load and preprocess the Adult Income dataset.<br>● Implement both Logistic Regression and Decision Trees.<br>● Compare the models based on metrics such as accuracy, precision, recall, and F1-score.<br>● Discuss the interpretability of both models and their suitability for the dataset. |
| 11 | Implement a Linear Support Vector Machine (SVM) to classify the Iris dataset. Visualize the decision boundary and discuss how the margin is determined.<br>Tasks:<br>● Load and preprocess the Iris dataset.<br>● Implement a Linear SVM for binary classification (e.g., classify Setosa vs. Non-Setosa).<br>● Visualize the decision boundary and margin.<br>● Discuss the concept of the margin and how it influences classification. |
| 12 | Implement and compare the performance of SVM classifiers with linear, polynomial, and RBF kernels on the Fashion MNIST dataset. Analyze the advantages and disadvantages of each kernel type. |

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| | Tasks:<br>● Load and preprocess the Fashion MNIST dataset.<br>● Implement SVM with linear, polynomial, and RBF kernels.<br>● Compare the classification performance for each kernel.<br>● Discuss the strengths and weaknesses of each kernel type. |
| 13 | Implement and train a Multilayer Feed-Forward Network (MLP) on the Wine Quality dataset. Experiment with different numbers of hidden layers and neurons, and discuss how these choices affect the network’s performance.<br>Tasks:<br>● Load and preprocess the Wine Quality dataset.<br>● Design and implement an MLP with varying architectures (different hidden layers and neurons).<br>● Train and evaluate the network.<br>● Discuss the impact of architecture choices on performance. |
| 14 | Implement and compare the performance of a neural network using different activation functions (Sigmoid, ReLU, Tanh) on the MNIST dataset. Analyze how each activation function affects the training process and classification accuracy.<br>Tasks:<br>● Load and preprocess the MNIST dataset.<br>● Implement neural networks using Sigmoid, ReLU, and Tanh activation functions.<br>● Train and evaluate each network.<br>● Compare training times, convergence, and classification accuracy. |
| 15 | Implement and perform hyperparameter tuning for a neural network on the Fashion MNIST dataset. Experiment with different learning rates, batch sizes, and epochs, and discuss the impact on model performance.<br>Tasks:<br>● Load and preprocess the Fashion MNIST dataset.<br>● Experiment with different hyperparameters (learning rate, batch size, epochs).<br>● Train and evaluate the network.<br>● Discuss how hyperparameter choices affect model performance. |
| 16 | Implement and compare hierarchical (agglomerative) and partitional (K-means) clustering algorithms on the Mall Customers dataset. Discuss the strengths and weaknesses of each method based on clustering results and evaluation metrics.<br>Tasks:<br>● Load and preprocess the Mall Customers dataset.<br>● Apply both hierarchical (agglomerative) and K-means clustering.<br>● Compare results using metrics such as inertia, silhouette score, and clustering |

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| | visualization.<br>● Discuss the advantages and disadvantages of each clustering method. |
| 17 | Implement and apply K-means clustering to the Digits dataset. Experiment with different numbers of clusters and evaluate the clustering results using metrics such as inertia and silhouette score. Analyze how the choice of K affects clustering performance.<br>Tasks:<br>● Load and preprocess the Digits dataset.<br>● Implement K-means clustering with various numbers of clusters.<br>● Evaluate clustering performance using inertia and silhouette score.<br>● Analyze the impact of the number of clusters on clustering quality. |
| 18 | Implement bootstrapping and cross-validation on the Iris dataset. Compare the model performance metrics (e.g., accuracy, F1-score) obtained using these resampling methods. Discuss the advantages and disadvantages of each method.<br>Tasks:<br>● Load and preprocess the Iris dataset.<br>● Implement bootstrapping to generate multiple samples and evaluate the model.<br>● Implement k-fold cross-validation and evaluate the model.<br>● Compare the performance metrics and discuss the pros and cons of each resampling method. |
| 19 | Implement bagging and boosting ensemble methods on the Titanic dataset. Compare the performance of both methods in terms of accuracy, precision, recall, and F1-score. Discuss how each method improves model performance and their respective strengths and weaknesses.<br>Tasks:<br>● Load and preprocess the Titanic dataset.<br>● Implement bagging using a base classifier (e.g., decision tree) and evaluate performance.<br>● Implement boosting using a boosting algorithm (e.g., AdaBoost) and evaluate performance.<br>● Compare performance metrics and discuss the strengths and weaknesses of each method. |
| 20 | Investigate the bias-variance tradeoff using polynomial regression on the Boston Housing dataset. Plot the training and validation errors for various polynomial degrees and discuss the tradeoff between bias and variance.<br>Tasks:<br>● Load and preprocess the Boston Housing dataset.<br>● Implement polynomial regression with varying degrees. |

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| | ● Plot training and validation errors for each degree.<br>● Discuss the bias-variance tradeoff and its impact on model performance. |